NEWTON SCHOOL · ANSWER KEY

PRE-CALCULUS | 🔑 ANSWER KEY

Sequences & Series: Arithmetic, Geometric & Infinite

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Standards: *CCSS.MATH.CONTENT.HSF-BF.A.2 · HSF-IF.A.3*

📐 **Formulas Used**

Type	nth term	Sum
Arithmetic	$t_n = t_1 + (n-1)d$	$S_n = \frac{n}{2}(t_1 + t_n)$
Geometric	$t_n = t_1 \cdot r^{n-1}$	$S_n = t_1 \cdot \frac{1-r^n}{1-r}$
Infinite Geom.	—	$S_\infty = \frac{t_1}{1-r} \quad (\,\lvert r\rvert < 1\,)$

Part A · Arithmetic Sequences

Problem 1 — Given $t_1 = 12, t_2 = 18$, find t_{20}.

Step 1. $d = t_2 - t_1 = 18 - 12 = 6$.

Step 2. $t_{20} = t_1 + (20-1)d = 12 + 19(6) = 12 + 114 = 126$.

$$\boxed{t_{20} = 126}$$

Problem 2 — Given $t_3 = 10, t_5 = 18$, find t_{40}.

Step 1. Two unknowns (t_1, d). Use the gap between t_3 and t_5 — that's 2 steps:

$t_5 - t_3 = 2d \;\Rightarrow\; 18 - 10 = 2d \;\Rightarrow\; d = 4$.

Step 2. Find t_1: $t_3 = t_1 + 2d \Rightarrow 10 = t_1 + 8 \Rightarrow t_1 = 2$.

Step 3. $t_{40} = t_1 + 39d = 2 + 39(4) = 2 + 156 = 158$.

$$\boxed{t_{40} = 158}$$

Problem 3 — Given $t_1 = 88, t_3 = 80$, find t_{50}.

Step 1. Gap from t_1 to t_3 is 2 steps:

$t_3 - t_1 = 2d \;\Rightarrow\; 80 - 88 = 2d \;\Rightarrow\; d = -4.$

Step 2. $t_{50} = t_1 + 49d = 88 + 49(-4) = 88 - 196 = -108.$

$$\boxed{t_{50} = -108}$$

Note: *negative d means the sequence decreases — and yes, t_{50} comes out negative.*

Part B · Geometric Sequences

Problem 4 — Given $t_1 = 5, t_2 = 10$, find t_{10}.

Step 1. $r = t_2/t_1 = 10/5 = 2$.

Step 2. $t_{10} = t_1 \cdot r^{10-1} = 5 \cdot 2^9 = 5 \cdot 512 = 2560$.

$$\boxed{t_{10} = 2560}$$

Problem 5 — Given $t_2 = 27, t_4 = 3$, find t_8.

Step 1. Two steps apart: $t_4/t_2 = r^2 = 3/27 = 1/9$.

$r = \pm\dfrac{1}{3}$ — both signs are valid; we'll see they give the same t_8.

Step 2. Find t_1 from $t_2 = t_1 \cdot r$:

Case $r = \frac{1}{3}$: $t_1 = 27/(1/3) = 81$.

Case $r = -\frac{1}{3}$: $t_1 = 27/(-1/3) = -81$.

Step 3. $t_8 = t_1 \cdot r^7$:

$r = \frac{1}{3}$: $t_8 = 81 \cdot (1/3)^7 = 81/2187 = 1/27$.

$r = -\frac{1}{3}$: $t_8 = -81 \cdot (-1/3)^7 = -81 \cdot (-1/2187) = 1/27$.

Both cases give the same answer (because $r^{2k} = (r^2)^k$ doesn't depend on r's sign for even-index gaps, and $t_8 = t_2 \cdot r^6 = 27 \cdot (1/9)^3 = 27/729 = 1/27$).

$$\boxed{t_8 = \frac{1}{27}}$$

Problem 6 — Given $t_1 = 2, t_2 = 2^{3/2}$, find t_{13}.

Step 1. $r = t_2/t_1 = 2^{3/2}/2 = 2^{3/2-1} = 2^{1/2}$.

Step 2. $t_{13} = t_1 \cdot r^{12} = 2 \cdot (2^{1/2})^{12} = 2 \cdot 2^6 = 2^7 = 128$.

$$\boxed{t_{13} = 128}$$

Why exponent rules win: $(2^{1/2})^{12} = 2^{(1/2)(12)} = 2^6$ — *much faster than computing* $\sqrt{2}^{12}$ *as a decimal.*

Part C · Series and Sums

Problem 7 — Find S_{10} of the arithmetic sequence: $5, 12, 19, 26, \ldots$

Step 1. Identify $t_1 = 5$, $d = 12 - 5 = 7$.

Step 2. Find $t_{10} = t_1 + 9d = 5 + 9(7) = 5 + 63 = 68$.

Step 3. Apply the sum formula $S_n = \frac{n}{2}(t_1 + t_n)$:

$S_{10} = \frac{10}{2}(5 + 68) = 5 \cdot 73 = 365.$

$$\boxed{S_{10} = 365}$$

Problem 8 — Find S_8 for the geometric sequence with $t_1 = 8, r = \frac{1}{2}$.

Step 1. Apply $S_n = t_1 \cdot \frac{1 - r^n}{1 - r}$:

$S_8 = 8 \cdot \frac{1 - (1/2)^8}{1 - 1/2} = 8 \cdot \frac{1 - 1/256}{1/2}$

Step 2. Simplify:

$= 8 \cdot \frac{255/256}{1/2} = 8 \cdot \frac{255}{256} \cdot 2 = \frac{16 \cdot 255}{256} = \frac{255}{16}$

Step 3. Decimal form: $255/16 = 15.9375$.

$$\boxed{S_8 = \frac{255}{16} = 15.9375}$$

Sanity check: $|r| = 1/2 < 1$, *so the partial sums converge to* $S_\infty = t_1/(1 - r) = 8/(1/2) = 16$. *After 8 terms we're at 15.9375 — within 0.0625 of the limit.* ✓

Part D · Infinite Geometric Series

Problem 9 — $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$

Step 1. Identify $t_1 = 1, r = 1/3$.

Step 2. Check $|r| = 1/3 < 1$ ✓ — series converges.

Step 3. Apply $S_\infty = \dfrac{t_1}{1-r}$:

$$S_\infty = \frac{1}{1-1/3} = \frac{1}{2/3} = \frac{3}{2}$$

$$\boxed{S_\infty = \frac{3}{2}}$$

Problem 10 — Convert the repeating decimal $0.\overline{45}$ to a fraction using an infinite geometric series.

Step 1. Write the decimal as a sum:

$$0.\overline{45} = 0.45 + 0.0045 + 0.000045 + \cdots$$
$$= \frac{45}{100} + \frac{45}{10000} + \frac{45}{1{,}000{,}000} + \cdots$$

Step 2. Identify $t_1 = \dfrac{45}{100}, r = \dfrac{1}{100}$. Check $|r| < 1$ ✓.

Step 3. Apply $S_\infty = \dfrac{t_1}{1-r}$:

$$S_\infty = \frac{45/100}{1-1/100} = \frac{45/100}{99/100} = \frac{45}{99}$$

Step 4. Simplify: $\dfrac{45}{99} = \dfrac{5}{11}$ (divide top & bottom by 9).

$$\boxed{0.\overline{45} = \frac{5}{11}}$$

Verify by long division: $5 \div 11 = 0.454545\ldots = 0.\overline{45}$ ✓

✅ All Boxed Answers

#	Type	Answer
1	Arithmetic	$t_{20} = 126$
2	Arithmetic	$t_{40} = 158$
3	Arithmetic	$t_{50} = -108$
4	Geometric	$t_{10} = 2560$
5	Geometric	$t_8 = 1/27$
6	Geometric	$t_{13} = 128$
7	Arithmetic Sum	$S_{10} = 365$
8	Geometric Sum	$S_8 = 255/16 = 15.9375$
9	Infinite Geom.	$S_\infty = 3/2$
10	Repeating decimal	$0.\overline{45} = 5/11$

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